

**University Interscholastic League
2022 – 2023 Elementary Number Sense Test**

B

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

Stop – Wait for Signal!

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| <p>(1) $6 \times 3 =$ _____</p> <p>(2) $314 + 413 =$ _____</p> <p>(3) $57 - 36 =$ _____</p> <p>(4) $3204 \div 4 =$ _____</p> <p>(5) $187 \div 11 =$ _____</p> <p>(6) $15 + 17 + 16 =$ _____</p> <p>(7) $8 \times 6 \times 2 =$ _____</p> <p>(8) $735 - 532 =$ _____</p> <p>(9) $71 \times 6 =$ _____</p> <p>*(10) $2301 + 1981 =$ _____</p> <p>(11) $11 \times 57 =$ _____</p> <p>(12) Which digit is in the hundreds place in 34976.15802? _____</p> <p>(13) $12 \times 16 =$ _____</p> <p>(14) 78630.4728 rounded to the ten-thousands place is _____</p> <p>(15) What is the remainder for $71513 \div 3$? _____</p> <p>(16) The number of even whole numbers between 9 and 20 is _____</p> <p>(17) $5 \times 10^3 + 4 \times 10^2 + 6 \times 10^{-2} =$ _____ (decimal)</p> <p>(18) $23 \times 6 - 23 \times 4 =$ _____</p> <p>(19) MMXXII = _____ (Arabic Numeral)</p> | <p>*(20) $667 \times 241 =$ _____</p> <p>(21) $24 - 4 \div 2 =$ _____</p> <p>(22) $8 + 12 + 16 + 20 + 24 =$ _____</p> <p>(23) $\frac{3}{5} =$ _____ percent</p> <p>(24) $2\frac{1}{2}$ days = _____ hours</p> <p>(25) $\frac{7}{18} + \frac{5}{18} =$ _____</p> <p>(26) $69 \times 71 =$ _____</p> <p>(27) 0.85 = _____ common fraction</p> <p>(28) If 42 ♥ costs 48¢ then 63 ♥ cost _____ ¢</p> <p>(29) $\frac{7}{18} \times 36 =$ _____</p> <p>*(30) $222 \times 361 =$ _____</p> <p>(31) 160 nickels = _____ dimes</p> <p>(32) The product of the two smallest primes greater than 40 is _____</p> <p>(33) 5 quarts = _____ ounces</p> <p>(34) $2250 \div 25 =$ _____</p> <p>(35) $11\frac{1}{9} \% =$ _____ common fraction</p> <p>(36) The LCM of 18 and 24 is _____</p> <p>(37) $93 \times 94 =$ _____</p> |
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- (38) The ratio in inches of 2 feet to 2 yards is _____
- (39) $3\frac{1}{5} \times 7\frac{1}{5} =$ _____ (mixed number)
- *(40) $376 \times 641 =$ _____
- (41) The side for a cube with volume 125 cm^3 is _____ cm
- (42) $23^2 =$ _____
- (43) The perimeter of a square with area 625 m^2 is _____ m
- (44) $2^3 + 2^2 =$ _____
- (45) $8\frac{3}{4} \div \frac{1}{4} =$ _____
- (46) $\sqrt{324} =$ _____
- (47) What is the perimeter of a right triangle with legs 12 and 5? _____
- (48) $22 \times 18 =$ _____
- (49) If $x = 25$, then $18 + 3x =$ _____
- *(50) $23 + 202 + 2021 + 2022 =$ _____
- (51) What is the number, **k**, in the sequence:
64, 32, 16, **k**, 4, 2, ...? _____
- (52) $10\frac{3}{8} - 5\frac{5}{8} =$ _____ (mixed number)
- (53) If the area of a circle is 324π , what is the diameter of the circle? _____
- (54) What is the volume of a rectangular box that measures 12" by 8" by 11"? _____ in^3
- (55) 123 (base 5) = _____ (base 10)
- (56) What whole number cubed minus twenty-five equals thirty-nine? _____
- (57) A triangle has sides of 18, 12, and a semiperimeter of 20. What is the third side? _____
- (58) If set $A = \{1, 3, 5, \dots, 11\}$ and set $B = \{3, 6, 9, 12\}$, then the number of elements in $A \cap B$ is _____
- (59) What is the perimeter of the square with a side length of $12\frac{1}{2}$? _____
- *(60) 281 days = _____ hours
- (61) 74 (base 8) = _____ (base 2)
- (62) $-3^4 \times (2) =$ _____
- (63) Two fair dice are thrown. What is the probability that the sum of the two sides showing is 3? _____
- (64) 1.5 square miles = _____ acres
- (65) $62^2 =$ _____
- (66) $6^6 \div 7$ has remainder of _____
- (67) How many edges does a rectangular box have? _____
- (68) If $3x + 12 < 36$, then $x <$ _____
- (69) $\frac{5}{8} + \frac{8}{5} = 2 +$ _____
- *(70) $111 \times \sqrt{325} =$ _____
- (71) The additive inverse of $-3\frac{1}{4}$ is _____
- (72) The area of a square with diagonal 16 is _____
- (73) If $7\frac{1}{2}\%$ of x is $22\frac{1}{2}\%$ of 6, then $x =$ _____
- (74) $(-24) \div (-8) \times (2) =$ _____
- (75) $22^2 + 11^2 =$ _____
- (76) $\left(18\frac{1}{2}\right)^2 - \left(11\frac{1}{2}\right)^2 =$ _____
- (77) What is the area of a trapezoid with bases 23, 18 and height 10? _____
- (78) $286 \times 49 =$ _____
- (79) $29^2 + 29$ _____
- *(80) $18 \times 20 \times 22 =$ _____

2022 – 2023 University Interscholastic League Elementary Number Sense Test B – Key

(1) 18	*(20) 152710 – 168784	(38) $\frac{1}{3}$	(59) 50
(2) 727	(21) 22		*(60) 6407 – 7081
(3) 21	(22) 80	(39) $23\frac{1}{25}$	(61) 111100
(4) 801	(23) 60		(62) -162
(5) 17	(24) 60	*(40) 228966 – 253066	(63) $\frac{1}{18}$
(6) 48	(25) $\frac{2}{3}$	(41) 5	(64) 960
(7) 96	(26) 4899	(42) 529	(65) 3844
(8) 203		(43) 100	(66) 1
(9) 426	(27) $\frac{17}{20}$	(44) 12	(67) 12
*(10) 4068 – 4496	(28) 72	(45) 35	(68) 8
(11) 627	(29) 14	(46) 18	(69) $\frac{9}{40}$
(12) 9	*(30) 76135 – 84149	(47) 30	
(13) 192	(31) 80	(48) 396	*(70) 1902 – 2101
(14) 80000	(32) 1763	*(50) 4055 – 4481	(71) $3\frac{1}{4}$; $\frac{13}{4}$; 3.25
(15) 2	(33) 160	(51) 8	(72) 128
(16) 5	(34) 90	(52) $4\frac{3}{4}$	(73) 18
(17) 5400.06	(35) $\frac{1}{9}$	(53) 36	(74) 6
(18) 46	(36) 72	(54) 1056	(75) 605
(19) 2022	(37) 8742	(55) 38	(76) 210
		(56) 4	(77) 205
		(57) 10	(78) 14014
		(58) 2	(79) 870
			*(80) 7524 – 8316

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.